1st Midterm · Unit 3 · Reduction of order · 5 points · 30 min · Standard exam level

P03 — A variable-coefficient equation with the solution x: reduction of order to x e^x, Abel's identity against the direct Wronskian, variation of parameters for the forcing x^4 and an IVP

Consider, for $x > 0$, the equation

$$x^2 y'' - x(x+2)\,y' + (x+2)\,y = g_0(x).$$

The homogeneous equation ($g_0 = 0$) has a solution that is a polynomial of degree one. In parts (c) and (d) the right-hand side is $g_0(x) = x^4$, and the initial data of (d) are

$$y(1) = \frac{1}{2}, \qquad y'(1) = -\frac{1}{2}.$$

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknown $y(x)$, independent variable x on the interval $(0, \infty)$.
- Operator: second order, **linear**, with **variable** coefficients; homogeneous in (a) and (b), inhomogeneous in (c) and (d).
- The leading coefficient x^2 vanishes at $x = 0$: $x = 0$ is a singular point, so all statements are made on $x > 0$, where the coefficients of the standard form are continuous and the existence and uniqueness theorem holds (Topic 3, Concept - Existence and Uniqueness Theorem for Second-Order ODEs).

Phase 2. Canonical formulation

Dividing by $x^2 \neq 0$, with $\frac{x+2}{x^2} = \frac{1}{x} + \frac{2}{x^2}$, the standard form $y'' + p(x)y' + q(x)y = g(x)$ is

$$y'' - \Big(1 + \frac{2}{x}\Big)y' + \Big(\frac{1}{x} + \frac{2}{x^2}\Big)y = \frac{g_0(x)}{x^2}, \qquad p(x) = -\Big(1 + \frac{2}{x}\Big), \quad g(x) = x^2 \text{ for } g_0 = x^4.$$

Part (a) · 1.5 points

First solution and reduction of order

Find the polynomial solution y_1 of degree one and, with $y = v\,y_1$, a second solution y_2 linearly independent of y_1.

Step 1. Polynomial solution

With $y = \alpha x + \beta$: $y' = \alpha$, $y'' = 0$, and

$$-x(x+2)\alpha + (x+2)(\alpha x + \beta) = (x+2)\big(-\alpha x + \alpha x + \beta\big) = \beta\,(x+2).$$

This vanishes for all x only if $\beta = 0$. So $y_1 = x$ (taking $\alpha = 1$).

Step 2. Substitution y = v x

By the product rule, $y = vx$ gives $y' = v'x + v$ and $y'' = v''x + v' + v' = v''x + 2v'$. Substituting:

$$x^2\big(xv'' + 2v'\big) - x(x+2)\big(xv' + v\big) + (x+2)\,xv.$$

Expanding term by term: $x^3v'' + 2x^2v' - (x^3 + 2x^2)v' - (x^2 + 2x)v + (x^2 + 2x)v = x^3v'' - x^3v'$. The terms in v cancel because y_1 solves the equation, as they must. Hence

$$x^3\big(v'' - v'\big) = 0 \iff v'' = v' \qquad (x > 0).$$

Step 3. First-order equation for w = v'

$w' = w$ is separable: $\frac{dw}{w} = dx$, $\ln|w| = x + C_1$, $w = Ce^x$. Integrating once more, $v = \int Ce^x\,dx = Ce^x + D$. Then $y = vx = C\,x\,e^x + D\,x$. The part Dx is the old solution; the new one is

$$y_2 = x\,e^x.$$

Check: $y_2' = (1+x)e^x$, $y_2'' = (2+x)e^x$; $x^2(2+x) - x(x+2)(1+x) + (x+2)x = x(x+2)\big[x - (1+x) + 1\big] = 0$, times e^x.

KEY RESULTS

- $y_1 = x$.
- $y = v\,x$ gives $x^3(v'' - v') = 0$, so $v' = Ce^x$ and $y_2 = x\,e^x$.
- General homogeneous solution $y_h = c_1 x + c_2\,x\,e^x$.

Part (b) · 1 point

Wronskian by Abel's identity and directly

Compute the Wronskian of $\{y_1, y_2\}$ from Abel's identity and from the determinant, and conclude that they form a fundamental set on $x > 0$.

Step 1. Abel's identity

With $p = -(1 + \frac{2}{x})$, $W' = -pW = \left(1 + \frac{2}{x}\right)W$. Separating, $\frac{dW}{W} = \left(1 + \frac{2}{x}\right)dx$, and with the antiderivative $x + 2\ln x$ ($x > 0$):

$$\ln|W| = x + 2\ln x + C_1 \implies W(x) = C\,x^2 e^x.$$

Step 2. Direct determinant

$$W = \det\begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} = \det\begin{pmatrix} x & x\,e^x \\ 1 & (1+x)e^x \end{pmatrix} = x(1+x)e^x - x\,e^x \cdot 1 = x^2 e^x.$$

The two results agree with $C = 1$.

Step 3. Conclusion

$x^2 e^x > 0$ for $x > 0$, so y_1, y_2 are linearly independent there and $y_h = c_1 x + c_2 x e^x$ is the general solution of the homogeneous equation (Robinson Ch. 11). The dichotomy "either $W \equiv 0$ or W never vanishes" applies on intervals where p is continuous; $W(0) = 0$ happens at the singular point $x = 0$ and is no contradiction.

KEY RESULTS

- Abel: $W = C\,x^2 e^x$.
- Determinant: $W = x(1+x)e^x - x\,e^x = x^2 e^x$, so $C = 1$.
- $W \neq 0$ on $(0, \infty)$: fundamental set; $W(0) = 0$ only at the singular point.

Part (c) · 1.5 points

Variation of parameters for the forcing x^4

With $g_0(x) = x^4$, find a particular solution by variation of parameters and write the general solution.

Step 1. Formula

For $y'' + py' + qy = g$ with fundamental set $\{y_1, y_2\}$ (Robinson Eq. 18.8, Notes, p. 8):

$$y_p = -y_1 \int \frac{y_2\, g}{W}\, dx + y_2 \int \frac{y_1\, g}{W}\, dx, \qquad g = \frac{x^4}{x^2} = x^2.$$

Step 2. The two integrands

$$\frac{y_2\, g}{W} = \frac{x\, e^{x} \cdot x^2}{x^2 e^{x}} = x, \qquad \frac{y_1\, g}{W} = \frac{x \cdot x^2}{x^2 e^{x}} = x\, e^{-x}.$$

Step 3. The integrals

$\int x\, dx = \frac{x^2}{2}$. For the second, integrate by parts with $u = x$, $dv = e^{-x}dx$, so $du = dx$, $v = -e^{-x}$:

$$\int x\, e^{-x} dx = -x\, e^{-x} - \int (-e^{-x})\, dx = -x\, e^{-x} - e^{-x} = -(x+1)e^{-x}.$$

(Integration constants are omitted: they only add multiples of y_1 and y_2.)

Step 4. Particular and general solution

$$y_p = -x \cdot \frac{x^2}{2} + x\, e^{x} \cdot \big(-(x+1)e^{-x} \big) = -\frac{x^3}{2} - x^2 - x.$$

The term $-x$ is a multiple of y_1 and can be absorbed: $y_p = -\dfrac{x^3}{2} - x^2$. Check, with $y_p' = -\frac{3}{2}x^2 - 2x$, $y_p'' = -3x - 2$:

$$x^2(-3x-2) - x(x+2)\Big(-\frac{3}{2}x^2 - 2x\Big) + (x+2)\Big(-\frac{x^3}{2} - x^2\Big)$$

The products are $x(x+2)\big(\frac{3}{2}x^2 + 2x\big) = \frac{3}{2}x^4 + 2x^3 + 3x^3 + 4x^2 = \frac{3}{2}x^4 + 5x^3 + 4x^2$ and $(x+2)\big(-\frac{x^3}{2} - x^2\big) = -\frac{1}{2}x^4 - x^3 - x^3 - 2x^2 = -\frac{1}{2}x^4 - 2x^3 - 2x^2$, so

$$= \big(-3x^3 - 2x^2\big) + \Big(\frac{3}{2}x^4 + 5x^3 + 4x^2\Big) + \Big(-\frac{1}{2}x^4 - 2x^3 - 2x^2\Big) = x^4. \ \checkmark$$

The general solution is $y = c_1 x + c_2\, x\, e^{x} - \dfrac{x^3}{2} - x^2$.

KEY RESULTS

- $\displaystyle\int \frac{y_2\, g}{W}dx = \int x\, dx = \frac{x^2}{2}$, $\displaystyle\int \frac{y_1\, g}{W}dx = \int x\, e^{-x}dx = -(x+1)e^{-x}$.
- $y_p = -\dfrac{x^3}{2} - x^2$ (after dropping the multiple $-x$ of y_1).
- $y = c_1 x + c_2 x e^{x} - \dfrac{x^3}{2} - x^2$.

Part (d) · 1 point

Initial value problem

Solve the IVP with $y(1) = \frac{1}{2}$, $y'(1) = -\frac{1}{2}$ and give the largest interval on which the solution is guaranteed to exist and be unique.

Step 1. Conditions

$y(1) = c_1 + c_2 e - \frac{1}{2} - 1 = \frac{1}{2} \Rightarrow c_1 + e\, c_2 = 2$. $y'(1) = c_1 + 2e\, c_2 - \frac{3}{2} - 2 = -\frac{1}{2} \Rightarrow c_1 + 2e\, c_2 = 3$.

Step 2. Constants

Subtracting the first equation from the second: $e\, c_2 = 1$, $c_2 = e^{-1}$; then $c_1 = 2 - 1 = 1$. The system is uniquely solvable because its matrix is $\begin{pmatrix} y_1(1) & y_2(1) \\ y_1'(1) & y_2'(1) \end{pmatrix} = \begin{pmatrix} 1 & e \\ 1 & 2e \end{pmatrix}$, whose determinant $2e - e = e$ is exactly $W(1) \neq 0$.

$$y(x) = x + x\, e^{x-1} - \frac{x^3}{2} - x^2.$$

Step 3. Interval

p, q and g of the standard form are continuous on $(0, \infty)$, which contains $x_0 = 1$; the existence and uniqueness theorem gives a unique solution on the whole of $(0, \infty)$. Nothing is guaranteed at or across $x = 0$.

KEY RESULTS

- $c_1 + e\, c_2 = 2$ and $c_1 + 2e\, c_2 = 3$, so $c_2 = e^{-1}$, $c_1 = 1$.
- $y(x) = x + x\, e^{x-1} - \dfrac{x^3}{2} - x^2$ on $(0, \infty)$.

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** as $x \to \infty$ the term $x\, e^{x-1}$ dominates and $y \to +\infty$; as $x \to 0^+$, $y \to 0$ and $y'(0^+) = 1 + e^{-1}$ (finite, though the theorem does not cover $x = 0$).
- **Checks:** y_1, y_2 and y_p were substituted back; Abel's W and the determinant agree; $y(1) = 1 + 1 - \frac{1}{2} - 1 = \frac{1}{2}$ and $y'(1) = 1 + 2 - \frac{3}{2} - 2 = -\frac{1}{2}$.
- **Common mistakes in the exam:** (1) applying variation of parameters to x^4 without dividing by the leading coefficient x^2; (2) taking $p = -(x+2)$ instead of $-(x+2)/x$ in Abel's identity; (3) a sign slip in $\int x e^{-x} dx$; (4) keeping $-x$ in y_p and still reporting $c_1 = 1$: with $y_p = -\frac{x^3}{2} - x^2 - x$ the IVP gives $c_1 = 2$, and the solution $y(x)$ is the same.